

Water-scene classification under a 100-label budget

EuroSAT100 (Sentinel-2, 13 bands, 64×64) · binary River ∪ SeaLake vs other classes · physics, tiny CNN, frozen ResNet-18

1. Problem

We ask a small-sample remote-sensing question: given one Sentinel-2 chip, is the scene a **water scene** or not? EuroSAT100 provides exactly 100 labelled 64×64 tiles with 13 spectral bands. We map the original ten land-cover classes to a binary label: **positive** if the official class is River or SeaLake (20 chips), **negative** otherwise (80 chips). Table 1 records the official 60/20/20 split. Only four water chips fall in validation and four in test, so a single accuracy number is not a sufficient summary.

This is **scene classification, not water segmentation**. Each tile has one label — the dominant land-cover class — not a pixel mask. A Highway chip that contains a river is still a **true negative** under these labels: the scene is a road scene, even if water is visible. A River chip still includes banks and vegetation. At 10 m GSD the tile covers 640 m × 640 m; mixed pixels are expected. We do not estimate flood extent or shoreline.

Split	Tiles	Water	Non-water	Positive rate
Train	60	12	48	0.20
Validation	20	4	16	0.20
Test	20	4	16	0.20
Total	100	20	80	0.20

Table 1. Official EuroSAT100 split used throughout. Source: `results/split_summary.json` in the repository.

Two facts drive the design. Water reflects in green and absorbs in the near-infrared and short-wave infrared, so spectral indices are a natural first model. And 80% of chips are non-water, so a classifier that always answers “no water” already reaches **80% accuracy**. Accuracy alone cannot be the headline metric.

2. Three-stage methodology

We compare three **separate** models on the same 100 chips. They are not fused, stacked, or voted at inference. Neural nets have to beat physics; they are not assumed to win. Cuts and hyperparameters are chosen on validation only. The test set is touched once. Thresholds come from unique validation scores (higher cut on ties). NDWI/MNDWI are used as **raw** scores, with no test-set scaling.

- Stage 1 — Physics (reference).** $NDWI = (B_{03} - B_{08}) / (B_{03} + B_{08})$ and $MNDWI = (B_{03} - B_{11}) / (B_{03} + B_{11})$. Each tile is summarised by mean, max, and wet-pixel fractions (eight features), then a regularised logistic regression (about ten coefficients) is fit. A one-parameter index threshold is also reported. *Motivation:* with 60 training images the safest model is few parameters plus a real spectral prior.
- Stage 2 — Tiny CNN from scratch.** Three conv–BN–ReLU blocks, global average pooling, dropout, ~24–25k weights, class-weighted loss, early stopping on validation loss, no augmentation. *Motivation:* mean NDWI ignores shape (narrow river vs lake). This tests spatial learning under a tight budget. We call it capacity-reduced, not proven capacity-matched.
- Stage 3 — Frozen pretrained ResNet-18.** Sentinel-2 MoCo (SSL4EO-S12) backbone is frozen; a logistic head is fit on 512-D embeddings (~513 trainable parameters). Unused bands are zero-masked so every band subset uses the same stem. *Motivation:* fine-tuning eleven million weights on 60 labels is not a reasonable estimator. Unlabelled pretraining is allowed; extra labelled EuroSAT images are not.



Figure 1. Train examples (2–98% display stretch). Top: true colour ($B_{04}-B_{03}-B_{02}$). Bottom: water composite (SWIR1–NIR–green). SeaLake is dark in the composite because water absorbs NIR/SWIR; the river stays dark while vegetation is bright. File: `results/fig_examples.png`.

3. Evaluation metrics

The class prior is 0.80 non-water. We keep accuracy as a sanity check against that floor and put the weight on metrics that treat water as the positive class.

Accuracy — *pro*: easy to explain; anything below 0.80 is worse than always predicting non-water. *con*: dominated by the negative class. **Precision** — *pro*: penalises false water calls. *con*: can look high if the model is timid. **Recall** — *pro*: how many water scenes we catch. *con*: one miss among four positives moves recall by 0.25; a Wilson 95% interval on 3/4 is about 0.30–0.95 (Table 2). **F1** — *pro*: balances precision and recall at one operating point. *con*: depends on the validation cut, which is noisy with four val positives. **ROC-AUC** — *pro*: threshold-free ranking. *con*: optimistic with 80% negatives. **PR-AUC (AP)** — *pro*: ranking for the positive class; chance equals 0.20. *con*: still has no operating point. Several models in Table 2 have AP = 1.0 and F1 = 0.86: they **order** the twenty chips correctly but the **val-chosen cut** misses one water scene.

Model	Acc.	Acc. 95% CI	F1	AP	Recall 95% CI
Majority (always non-water)	0.80	0.58–0.92	0.00	0.20	0.00–0.49
NDWI threshold	0.95	0.76–0.99	0.86	0.89	0.30–0.95
Logistic, NDWI/MNDWI stats	0.95	0.76–0.99	0.86	1.00	0.30–0.95
Tiny CNN, water bands	1.00	0.84–1.00	1.00	1.00	0.51–1.00
Frozen probe, RGB only	0.80	0.58–0.92	0.33	0.65	0.05–0.70
Frozen probe, water bands	0.95	0.76–0.99	0.86	1.00	0.30–0.95
Frozen probe, all 13 bands	1.00	0.84–1.00	1.00	1.00	0.51–1.00

Table 2. Official 20-image test set (4 water). AP is average precision (PR-AUC). Intervals are Wilson 95% CIs on accuracy and recall. Source: results/official_metrics.md. Full band-ablation and random-4 arms are not repeated here; they are in the same file on GitHub.

4. Results

Table 2 is the official split. Majority accuracy is 0.80 with F1 = 0, as expected. A raw NDWI threshold already reaches F1 = 0.86. Logistic regression on the eight index statistics reaches AP = 1.0 and F1 = 0.86: perfect ranking, one threshold error. The tiny CNN on water bands and the all-band frozen probe can look perfect on *this* 20-chip split (F1 = 1.0). RGB-only embeddings do not (F1 = 0.33). Those perfect scores sit inside wide intervals (recall 0.51–1.00 for 4/4). Pairwise McNemar tests on the same twenty chips are not significant (all $p \geq 0.125$); with four positives they cannot establish a winner. The McNemar table is omitted here and is stored as results/mcnemar.json.

Figure 2 and Table 3 report stratified five-fold cross-validation on the **same** 100 chips — a stability check, not extra labelled data. This is the more honest summary. Logistic index features give F1 0.80 ± 0.17 . The frozen probe is in the same band (water bands 0.83 ± 0.13 ; all bands 0.77 ± 0.21). The tiny CNN is worse and much noisier (0.63 ± 0.34). Neural complexity did not buy stability.

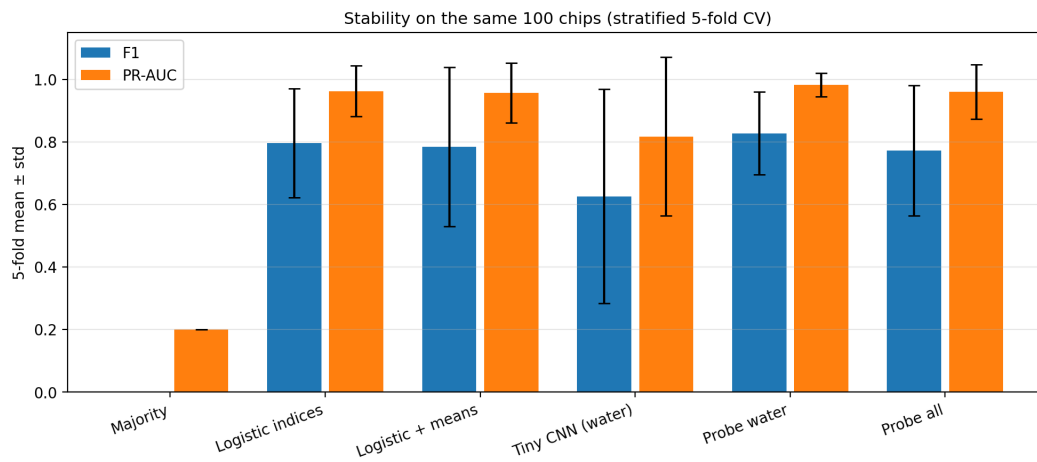


Figure 2. Stratified 5-fold F1 and PR-AUC (mean ± std) on the same 100 chips. The tiny CNN has the largest error bar. Physics logistic and the frozen probe sit together near $F1 \approx 0.8$. File: `results/fig_cv_stability.png`.

Model	CV F1 (mean ± sd)	CV AP (mean ± sd)
Majority	0.00 ± 0.00	0.20 ± 0.00
Logistic, index statistics	0.80 ± 0.17	0.96 ± 0.08
Tiny CNN, water bands	0.63 ± 0.34	0.82 ± 0.25
Frozen probe, water bands	0.83 ± 0.13	0.98 ± 0.04
Frozen probe, all 13 bands	0.77 ± 0.21	0.96 ± 0.09

Table 3. Supplementary 5-fold CV on all 100 chips (inner validation for thresholds). Source: `results/cv_f1.json`.

5. Reading

We treated this as a small-sample remote-sensing study, not an accuracy contest. Physics first, because the spectrum of water is known; a small CNN next, to test spatial learning; a frozen Sentinel-2 ResNet last, to use pretraining without fitting millions of weights. Accuracy is checked against the 0.80 floor in Table 2. F1 and AP are the headlines and are kept separate because one is a cut and the other is a ranking. The evidence in Tables 2–3 and Figure 2 is that **spectral features are a competitive reference**, that a tiny CNN is not shown to be reliable at this n, and that four test positives cannot support a superiority claim. If more labels arrive, the same ladder still applies: keep the index model, keep the frozen probe, and only then unfreeze the backbone.

Remaining artefacts (full official-split dump, band-ablation bars, PR/ROC curves, confusion matrices, training curves, per-band reflectance table, error galleries) are not reproduced here. They are on GitHub: https://github.com/pranavtripathi6844/thesis_assignment- in the `results/` folder.